

# EXP → V5 Cheat Sheet

*Pull this out when you join the team*

Everything you learned on EXP, in the words the V5 team uses. A few names change; the thinking does not. **A VEX match = an autonomous period + a driver period — and you already do both (your Sumo auto-hunt and your tank driving).**

| Concept                                | A&R · VEX EXP (VEXcode)                    | Team · VEX V5 (SpartanDrive)            |
|----------------------------------------|--------------------------------------------|-----------------------------------------|
| <b>Robot's computer</b>                | EXP Brain — built-in inertial sensor       | V5 Brain — built-in IMU                 |
| <b>The motors</b>                      | Smart Motor — 200 RPM, fixed               | Smart Motor — swappable cartridges      |
| <b>Bundle a side's motors</b>          | motor_group leftSide = motor_group(...)    | the left/right groups SpartanDrive uses |
| <b>Whole-robot drive object</b>        | a (smart) drivetrain                       | the drive object                        |
| <b>Tank (two-stick) driving</b>        | left & right motor_group from the sticks   | drive.tank(leftY, rightY)               |
| <b>Arcade (one-stick) driving</b>      | Drivetrain.arcade(power, turn)             | drive.split_arcade(leftY, rightX)       |
| <b>Drive a set distance</b>            | Drivetrain.driveFor(forward, 24, inches)   | drive.drive_distance(24)                |
| <b>Turn to a heading (absolute)</b>    | Drivetrain.turnToHeading(90, degrees)      | drive.turn_to_angle(90)                 |
| <b>Turn a set amount (relative)</b>    | Drivetrain.turnFor(right, 90, degrees)     | drive.turn_to_angle(heading + 90)       |
| <b>Which way am I facing?</b>          | Brain inertial → heading                   | drive.heading()                         |
| <b>Where am I on the field?</b>        | — not tracked on EXP —                     | odometry → drive.get_pose()             |
| <b>Ready the sensor</b>                | inertial calibrates at Brain startup       | drive.calibrate()                       |
| <b>Run-by-itself routine</b>           | your Sumo auto-hunt                        | autonomous()                            |
| <b>Driver loop (1 drive call/pass)</b> | while (true)<br>{ Drivetrain.arcade(...) } | opcontrol() { ... drive.tank(...) }     |
| <b>Where you write &amp; run code</b>  | VEXcode (VS Code) → Download & Run         | PROS — different editor & layout        |
| <b>The C++ language &amp; logic</b>    | same thinking                              | same thinking                           |

## On the team — going further (SpartanDrive)

| On the team you'll also meet      | What it does                                                      |
|-----------------------------------|-------------------------------------------------------------------|
| drive.swing_turn(90, true)        | Pivot on one side; the other holds — a wide, sweeping turn.       |
| drive.drive_distance_chain(24, 3) | Drive, but exit early (still rolling) to flow into the next move. |
| drive.turn_to_angle_chain(90, 5)  | Turn, but exit early to chain into the next move.                 |

| On the team you'll also meet                                | What it does                                   |
|-------------------------------------------------------------|------------------------------------------------|
| drive.get_pose() · drive.set_pose(...)                      | Read or set the robot's (x, y) field position. |
| drive.reset_odom()                                          | Zero the odometry position.                    |
| drive.drive_raw(left, right)                                | Raw power to each side — the escape hatch.     |
| drive.set_drive_constants(...) ·<br>set_turn_constants(...) | Tune the drive / turn PID gains.               |
| drive.stop() · drive.hottest_motor_temp()                   | Stop the drive · check the hottest motor.      |

**The one line:** same C++ thinking, same robot concepts — a handful of methods get renamed and the editor changes when you reach the team. You're not starting over; you're picking up where this course leaves off.